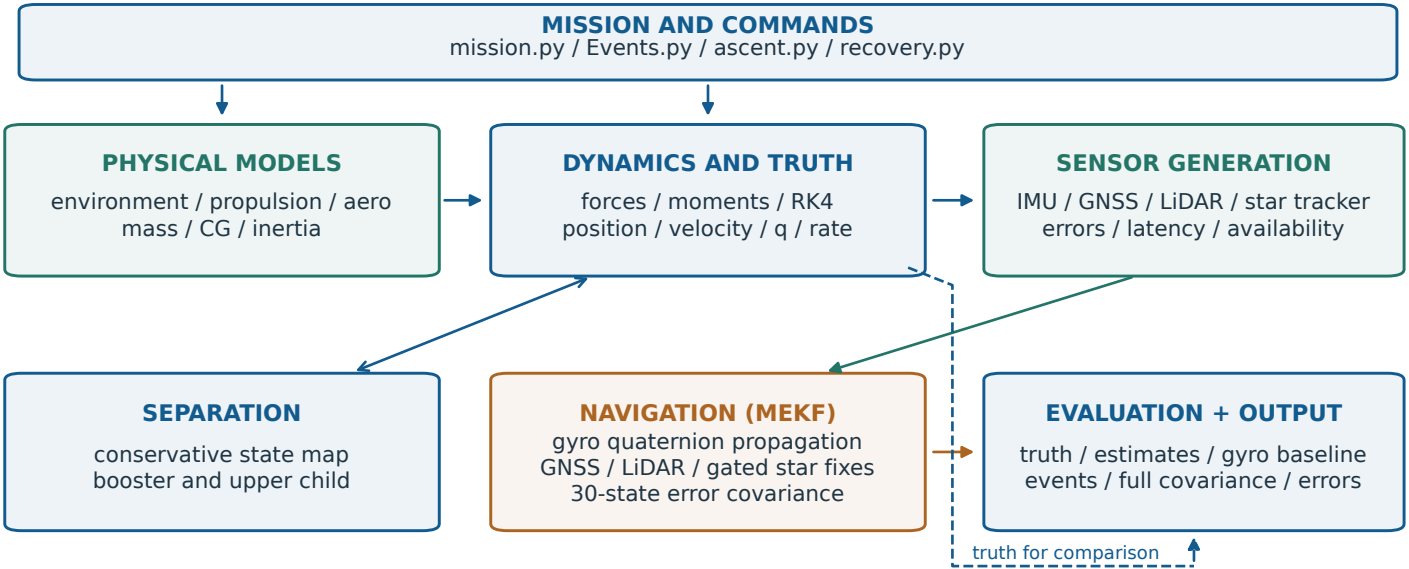
MISSION AND COMMANDS
mission.py / Events.py / ascent.py / recovery.py
PHYSICAL MODELS
environment / propulsion / aero
mass / CG / inertia
DYNAMICS AND TRUTH
forces / moments / RK4
position / velocity / q / rate
SENSOR GENERATION
IMU / GNSS / LiDAR / star tracker
errors / latency / availability
SEPARATION
conservative state map
booster and upper child
NAVIGATION (MEKF)
gyro quaternion propagation
GNSS / LiDAR / gated star fixes
30-state error covariance
EVALUATION + OUTPUT
truth / estimates / gyro baseline
events / full covariance / errors
truth for comparison